

Cryptoball

Protocol POC Requirements & Architecture

Devnet MVP · Solana · Anchor 0.32.1

Draft v1 · 2026-10-01 · defaults D1–D6 await captain sign-off

Method: Turbin3 architecture-diagramming process (requirements → actors → state ownership → reference table → diagrams → MVP cut → self-check → red-team).

Format follows the QUSD “Protocol POC Requirements” example: atomic requirements, overview, one flow per state change, end-to-end overview.

Protocol POC Requirements

Each line is one action. R-xx IDs appear on every diagram element. Machine-checked: 87 requirements ↔ 38 elements, 0 failures.

A. Config & authority

- R-01 The protocol shall create one global Config account (seed "config") on initialize.
- R-02 The protocol shall reject initialize unless it is signed by the program upgrade authority.
- R-03 The protocol shall store the treasury wallet address in Config at initialize.
- R-04 The protocol shall store the entry-token mint (USDC) in Config at initialize.
- R-05 The protocol shall store the fee in basis points in Config at initialize.
- R-06 The protocol shall reject any fee above MAX_FEE_BPS whenever the fee is set.
- R-07 The protocol shall allow the admin to update the fee.
- R-08 The protocol shall allow the admin to update the treasury address.
- R-09 The protocol shall allow the admin to pause ticket sales.
- R-10 The protocol shall allow the admin to resume ticket sales.
- R-11 The protocol shall allow the admin to nominate a new admin.
- R-12 The protocol shall require the nominee's signature to accept the admin role.
- R-76 The protocol shall emit an event on every Config mutation.

B. Campaign

- R-13 The protocol shall allow the admin to create a campaign with an id, ticket price, close timestamp and max tickets.
- R-14 The protocol shall reject campaign creation when the close timestamp is not in the future.
- R-15 The protocol shall reject campaign creation when the ticket price is zero.
- R-16 The protocol shall reject campaign creation when max tickets is zero or above MAX_TICKETS.
- R-17 The protocol shall create the campaign vault as the associated token account of the Campaign PDA for the configured mint.
- R-18 The protocol shall create one Metaplex Core collection per campaign via CPI.
- R-19 The protocol shall initialize the campaign in state Open with ticket count zero.
- R-77 The protocol shall snapshot the Config fee into the Campaign at creation.
- R-20 The protocol shall expose no instruction that changes a campaign's price, fee or close time after creation.

C. Ticket purchase

- R-21 The protocol shall allow a buyer to purchase one ticket by supplying exactly five numbers.
- R-22 The protocol shall reject a purchase when the campaign is not Open.
- R-23 The protocol shall reject a purchase at or after the close timestamp.
- R-24 The protocol shall reject a purchase while sales are paused.
- R-25 The protocol shall reject any picked number outside 1..=69.
- R-26 The protocol shall reject picked numbers that are not strictly ascending.
- R-27 The protocol shall transfer the ticket price in USDC from the buyer to the campaign vault using transfer_checked.
- R-28 The protocol shall reject a purchase when the ticket count equals max tickets.
- R-29 The protocol shall create a Ticket account derived from the campaign and ticket index.
- R-30 The protocol shall store buyer, numbers, index and Core asset address in the Ticket.
- R-31 The protocol shall increment the campaign ticket count with checked arithmetic.

- R-32 The protocol shall mint one Core asset to the buyer with the numbers as an Attributes plugin.
- R-33 The protocol shall complete payment, Ticket creation and asset mint atomically in one transaction.
- R-34 The protocol shall emit a TicketPurchased event.
- R-35 The client shall generate quick-pick numbers from a cryptographically secure random source in ascending order.
- R-36 The client shall allow at most five cartons per checkout.

D. Draw (Switchboard)

- R-37 The protocol shall allow anyone to commit the draw only at or after the close timestamp.
- R-38 The protocol shall reject `commit_draw` when the campaign has zero tickets.
- R-39 The protocol shall reject a randomness account not owned by the Switchboard On-Demand program id pinned for the cluster.
- R-40 The protocol shall reject a randomness account whose seed slot is not the previous slot or that is already revealed.
- R-41 The protocol shall store the randomness account address and seed slot in the Campaign.
- R-42 The protocol shall move the Campaign from Open to DrawCommitted on `commit_draw`.
- R-43 The protocol shall allow anyone to settle only when the Campaign is DrawCommitted.
- R-44 The protocol shall reject settlement when the randomness account or seed slot differs from the stored values.
- R-45 The protocol shall obtain the 32-byte revealed value from the randomness account.
- R-78 The protocol shall reject settlement when the randomness has not yet been revealed.
- R-46 The protocol shall derive the winning ticket index as the revealed value (u128) modulo the ticket count.
- R-47 The protocol shall reject settlement unless the supplied Ticket account is the PDA for the derived index.
- R-48 The protocol shall compute the fee as pool times fee bps over 10,000, rounded down, with checked u128 math.
- R-49 The protocol shall transfer the fee from the vault to the treasury token account derived from Config.
- R-50 The protocol shall transfer the remaining pool from the vault to the winning buyer's token account.
- R-51 The protocol shall create the winning buyer's token account if it does not exist, paid by the settler.
- R-52 The protocol shall store the revealed value, winning index and winner in the Campaign.
- R-53 The protocol shall move the Campaign from DrawCommitted to Settled exactly once.
- R-54 The protocol shall emit a DrawSettled event.

E. Recovery

- R-55 The protocol shall allow anyone to cancel an Open campaign with zero tickets after its close timestamp.
- R-56 The protocol shall allow anyone to cancel a DrawCommitted campaign after the reveal timeout has elapsed.
- R-57 The protocol shall treat Settled and Cancelled as absorbing states.
- R-58 The protocol shall allow anyone to refund one Ticket's price to its stored buyer on a Cancelled campaign.
- R-59 The protocol shall mark a refunded Ticket Refunded.
- R-79 The protocol shall reject a second refund of the same Ticket.
- R-60 The protocol shall emit an event on campaign cancellation.
- R-80 The protocol shall emit an event on ticket refund.
- R-87 The protocol shall allow `commit_draw`, `settle_draw`, `cancel_campaign` and `refund_ticket` while sales are paused.

F. Client & transparency

- R-61 The protocol shall keep campaign, ticket, winner, randomness value and vault balance readable from public accounts without our API.
- R-62 The client shall list campaigns with their prize pool.

- R-81** The client shall show a countdown to each campaign's close time.
- R-63** The client shall let a user look up tickets by wallet address or purchase signature.
- R-64** The client shall show the winning ticket and its numbers on the results page.
- R-84** The client shall link the results page to the randomness account on the Explorer.
- R-65** The client shall connect a wallet through embedded social login or a wallet-adapter extension.
- R-66** The client shall show a devnet banner.
- R-82** The client shall block transactions when the wallet cluster is not devnet.
- R-67** The client shall refetch on-chain state after every confirmed transaction.
- R-68** The client shall present all user-facing text in Spanish (es-PE with es-419 fallback).
- R-83** The client shall format money, numbers and dates through Intl for the active Spanish locale.
- R-69** The client shall map every program error code to a Spanish message.
- R-70** [TBD-sponsor] The client shall let a buyer without SOL purchase by routing fees and rent through a sponsor fee payer.
- R-75** The keeper script shall call `commit_draw` once a campaign has closed.
- R-85** The keeper script shall call `settle_draw` once the randomness is revealed.
- R-86** The keeper script shall call `cancel_campaign` when a campaign is cancellable.

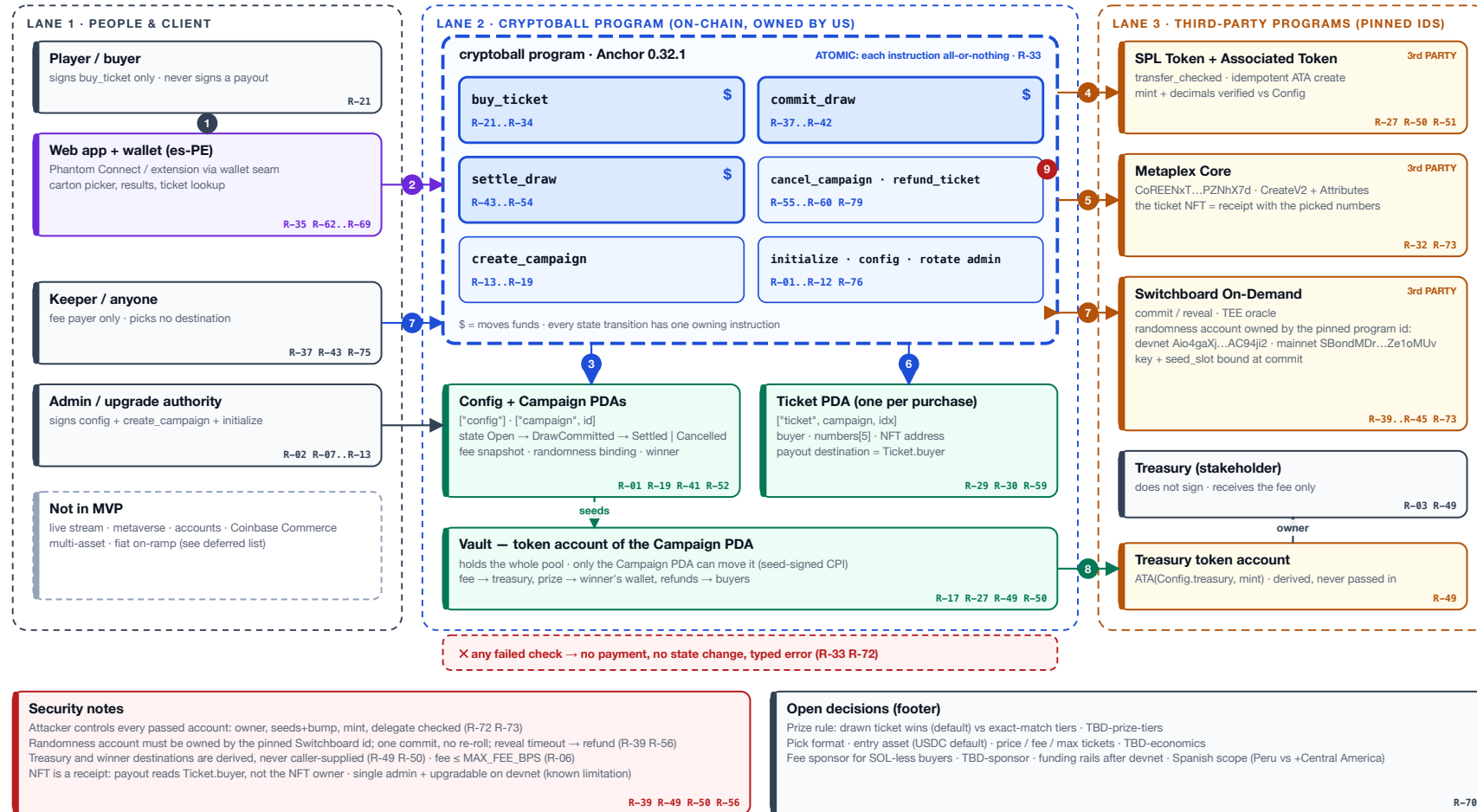
G. Non-functional

- R-71** The protocol shall use checked arithmetic on every money and counter path.
- R-72** The protocol shall return typed errors instead of panicking on user-controlled input.
- R-73** The protocol shall pin the Switchboard (per cluster), Core, token and associated-token program ids and never read them from instruction accounts.
- R-74** The system shall operate on devnet only in the MVP.

Overview

Cryptoball — Architecture overview (devnet MVP)

Three swimlanes · numbered arrows carry requirement IDs · thick outline = atomic boundary · red dashed = failure path (no state change)



Overview — how the flow works

1 – Player picks numbers

Five numbers 1–69, manual or quick-pick, up to five cartons. [R-35 R-36]

2 – Wallet signs buy_ticket

Phantom Connect or an extension wallet, behind a wallet seam. [R-21 R-65]

3 – Program validates

Open, before close, not paused, numbers valid, not sold out. [R-22..R-28]

4 – USDC moves to the vault

SPL token CPI, `transfer_checked`. [R-27]

5 – Ticket NFT minted

Metaplex Core, numbers as attributes. [R-32]

6 – Ticket PDA written

Steps 3–6 are one atomic transaction. [R-29..R-34]

7 – Draw committed

After close, bound to a Switchboard randomness account. [R-37..R-42]

8 – Settlement and automatic payout

Fee to treasury, prize to the winner's own wallet. [R-43..R-54]

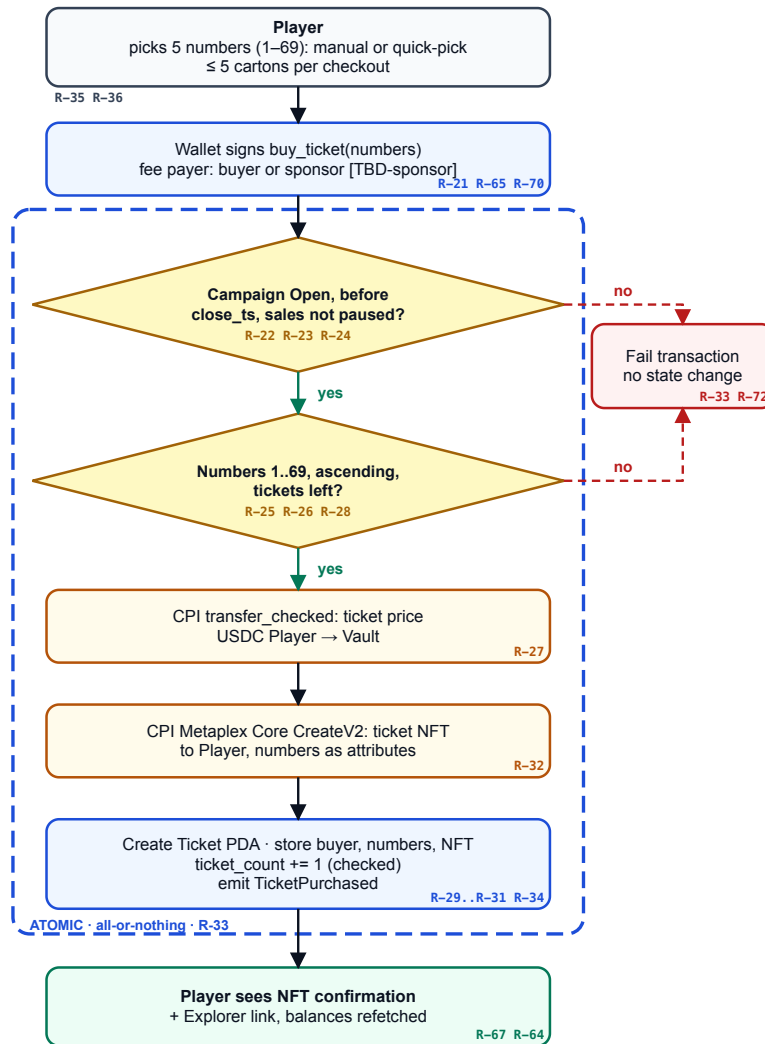
9 – Recovery

Cancel and refund if no tickets or the reveal times out. [R-55..R-60]

Not in the MVP: live-streamed drawings, metaverse gamification, account system / emails, Coinbase Commerce rail, multi-asset campaigns, wallet notification panel, fiat on-ramp in the app, exact-match prize tiers, mainnet.

Actors and signers. Player: signs `buy_ticket` only. Admin: signs admin instructions; nominee signs `accept_admin`. Upgrade authority: signs `initialize` only. Keeper / anyone: fee payer only, chooses no destination. Treasury: does not sign. Switchboard oracle (TEE): third party, writes the revealed randomness. Metaplex Core, SPL Token / ATA, Switchboard programs: external, ids pinned.

Buy ticket



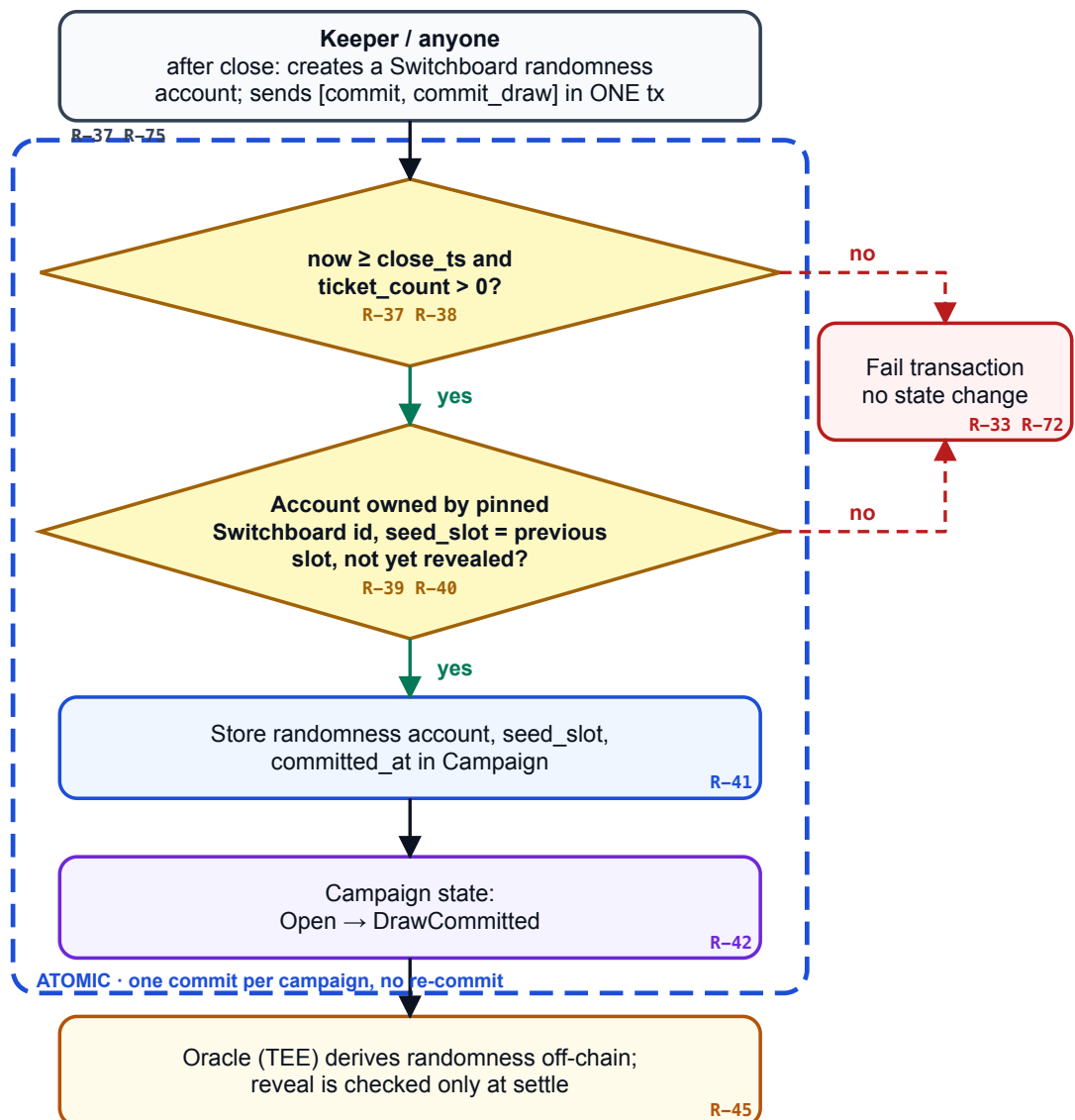
The player picks five numbers between 1 and 69, by hand or with quick-pick, up to five cartons per checkout (R-35 R-36). The wallet signs `buy_ticket`.

The program checks that the campaign is open, before its close time, sales are not paused, the numbers are valid and ascending, and tickets are left (R-22 R-23 R-24 R-25 R-26 R-28).

Three things then happen in one atomic transaction: the ticket price in USDC moves from the player to the campaign vault (R-27), a Metaplex Core NFT with the picked numbers as attributes is minted to the player (R-32), and a Ticket PDA stores buyer, numbers and NFT address (R-29 R-30). Any failure reverts all of it (R-33).

State: Ticket absent → Active; Campaign.ticket_count + 1. **Owner:** `buy_ticket`. **Accounts:** Config, Campaign, Ticket (new), Vault, buyer token account, Core asset (new).

Commit the draw



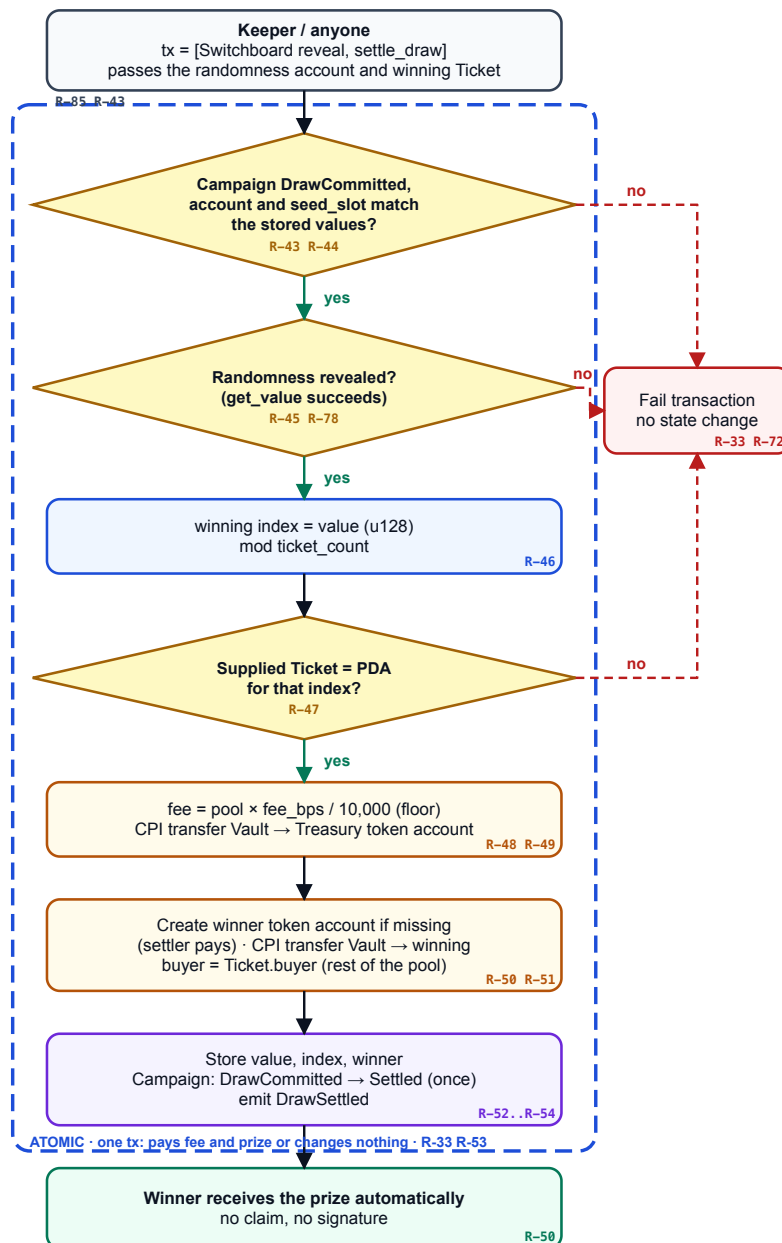
After sales close, anyone (normally the keeper script) creates a Switchboard randomness account and sends the Switchboard commit and `commit_draw` in one transaction (R-37 R-75).

The program rejects an empty campaign, an account not owned by the Switchboard program id pinned for the cluster, an account whose seed slot is not the previous slot, and one already revealed (R-38 R-39 R-40). It then stores the account address and seed slot and moves the campaign to **DrawCommitted** (R-41 R-42).

There is no re-commit instruction, so nobody can re-roll the draw. Sales were already closed, so nobody can buy after seeing the result.

State: Open → DrawCommitted. **Owner:** `commit_draw`.

Settle the draw and pay the winner

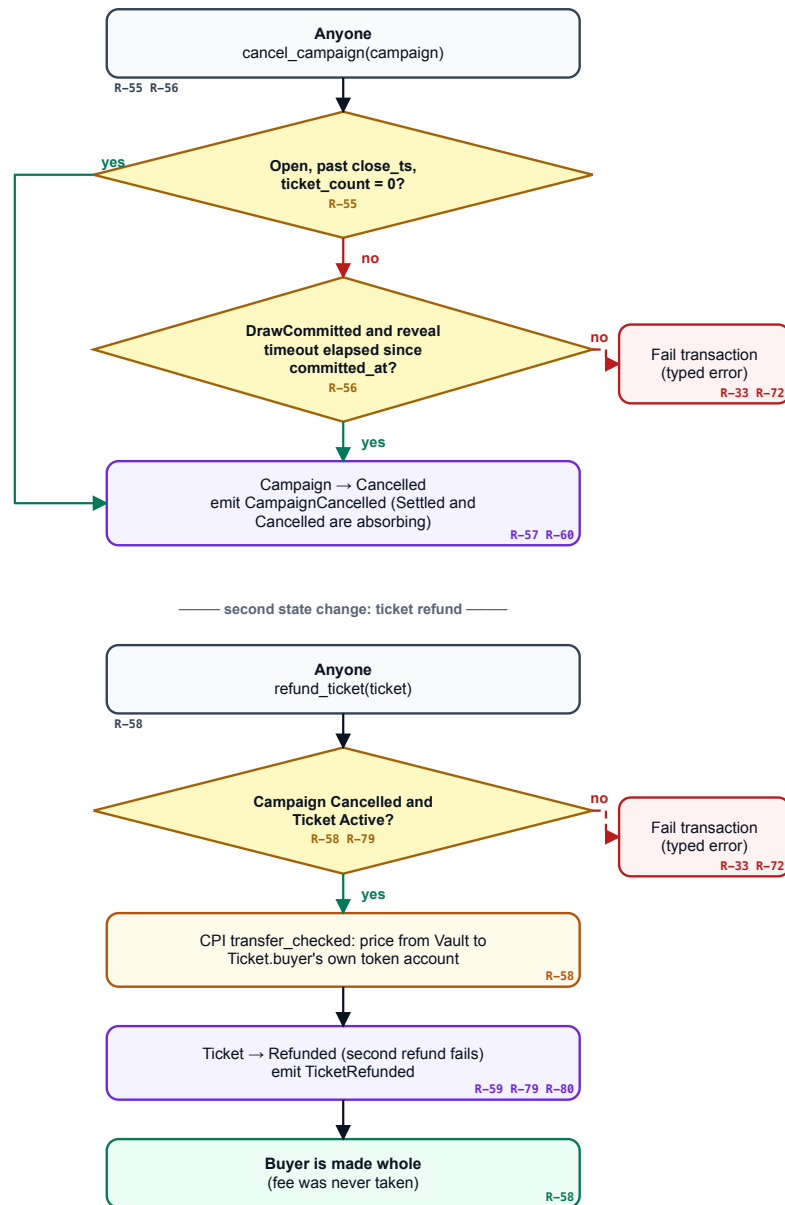


Anyone sends `reveal` and `settle_draw` in one transaction. The program checks that the campaign is `DrawCommitted` and that the randomness account and seed slot match the stored ones (R-43 R-44), and that the value is revealed (R-45 R-78).

The winning ticket index is the 128-bit revealed value modulo the ticket count (R-46); the supplied Ticket account must be the PDA for that index (R-47). The fee ($\text{price} \times \text{fee bps} \div 10,000$, rounded down) goes to the treasury token account derived from Config (R-48 R-49); the rest of the pool goes to the buyer stored in the winning Ticket, creating their token account if needed (R-50 R-51).

The winner does nothing: no claim, no signature. **State:** `DrawCommitted` → `Settled (once)`; winner, value and index stored (R-52 R-53). **Default winner rule (D2):** one drawn ticket wins; picked numbers are recorded on the NFT only. **Owner:** `settle_draw`.

Cancel a campaign and refund tickets

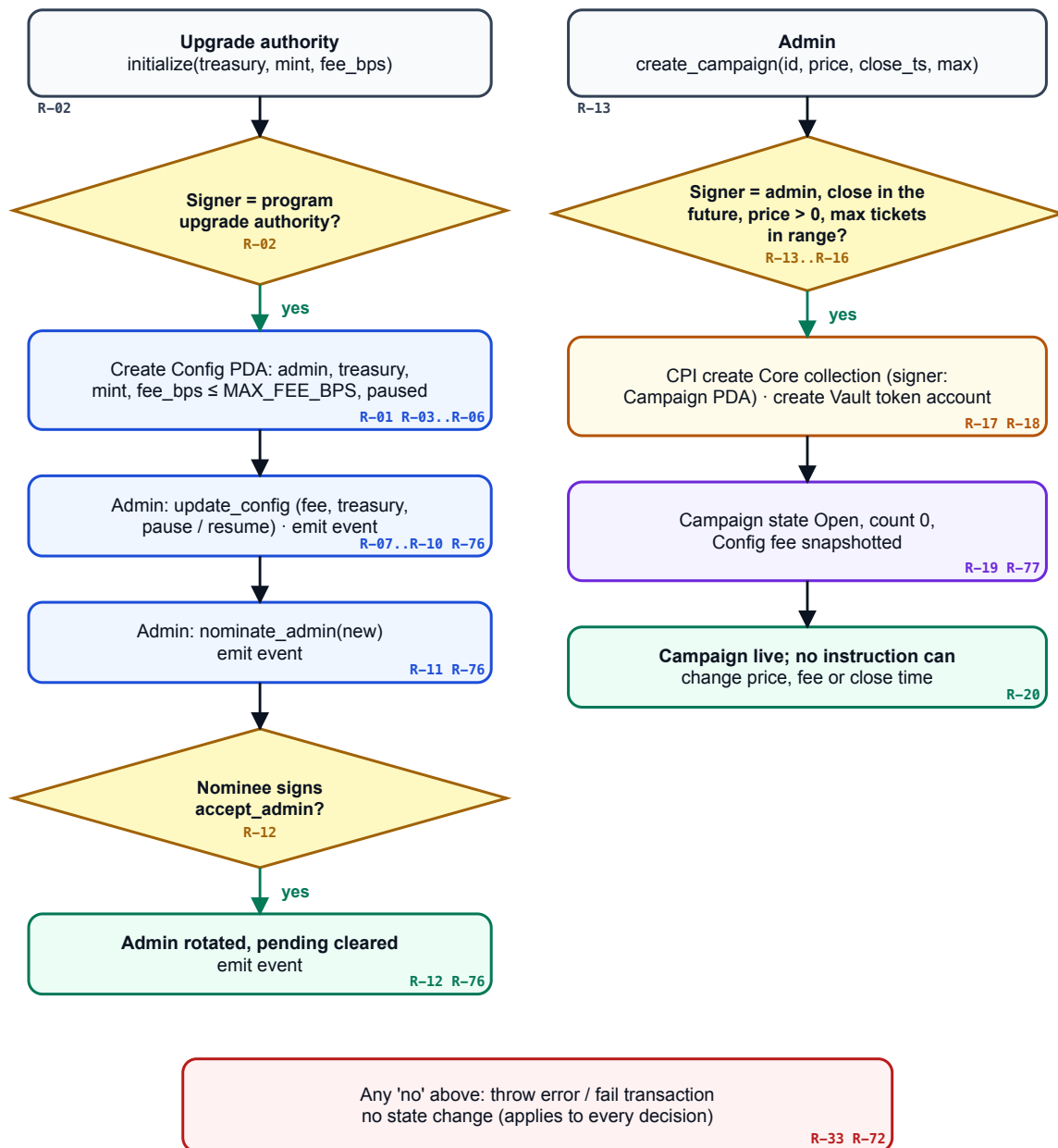


Cancel. Anyone can cancel an open campaign that closed with zero tickets (R-55), or a DrawCommitted campaign whose reveal never completed within the timeout (R-56). Settled and Cancelled are absorbing (R-57).

Refund. On a cancelled campaign anyone can refund one ticket: the full price goes from the vault to the buyer stored in the Ticket, once (R-58 R-59 R-79). The caller never chooses the destination.

Both work while sales are paused, so funds are never trapped (R-87). **Owners:** `cancel_campaign`, `refund_ticket`.

Set up: initialize, configure and create a campaign



Initialize is allowed only for the program upgrade authority and creates the single Config PDA (R-01 R-02). The fee is capped by a constant (R-06). The admin can update the fee, treasury and pause flag (R-07..R-10); admin rotation takes two steps and the nominee must sign (R-11 R-12). Every change emits an event (R-76).

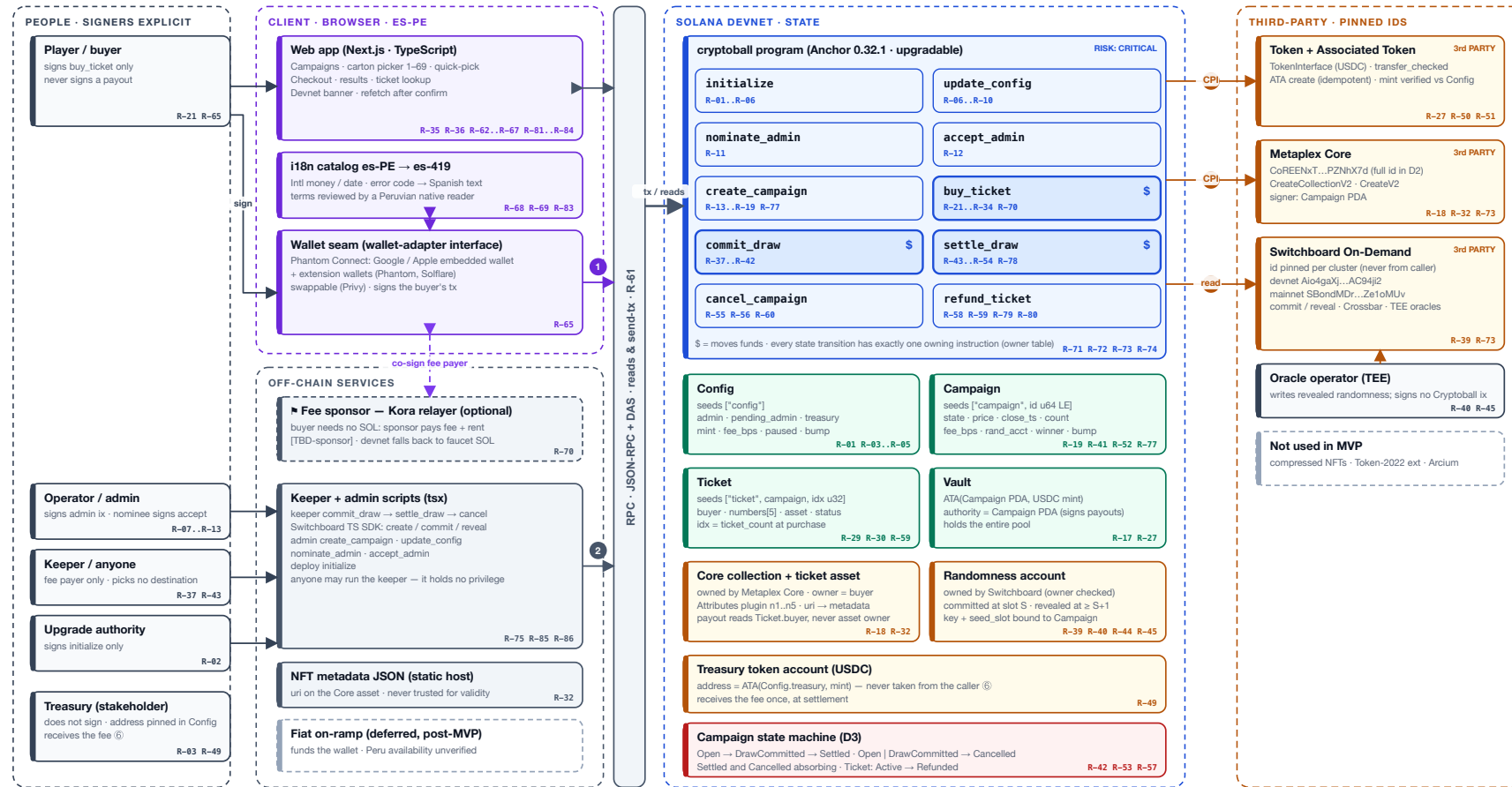
Create campaign (admin): validates the close time, price and max tickets (R-14..R-16), creates the Core collection and the vault (R-17 R-18), starts the campaign Open with count zero and a snapshot of the fee (R-19 R-77). No instruction can change a campaign's price, fee or close time afterwards (R-20).

Owners: `initialize`, `update_config`, `nominate_admin`, `accept_admin`, `create_campaign`.

End-to-end overview

Cryptoball — System Architecture (devnet MVP)

One transaction = one state change · circled numbers are explained under the diagram · 🏠 = default awaiting captain sign-off



Flows

- Player picks numbers → wallet signs buy_ticket (fee payer = sponsor, or buyer on devnet) → program pulls USDC into the Vault and mints the ticket NFT
- Keeper, after close: commit_draw (same tx as the Switchboard commit) → later reveal + settle_draw in one tx; cancel_campaign if the draw is stuck
- Token CPI: transfer_checked buyer → Vault on buy; Vault → treasury and Vault → winner on settle, signed by Campaign PDA seeds
- Core CPI: collection (create_campaign), asset (buy_ticket)
- Program reads the randomness account: owner = pinned Switchboard id, key and seed_slot equal the stored values, value revealed
- Treasury receives only the fee; its address always comes from Config

Legend

 Actor
 Client (browser)
 Off-chain service
 Instruction
 Program-owned account
 External / third-party
 State machine
 Deferred (post-MVP)
 → call / CPI
 - -> optional / read / off-chain
 R-xx = requirement id

Account map, PDA seeds and ownership

D2 — Account map, PDA seeds and ownership

Every account: owner program · seeds · fields · who may mutate it. Bumps are stored at init and re-verified; never recomputed on-chain.



Rent: Config, Campaign and Ticket rent is paid by the payer (admin, or buyer / sponsor). Per-purchase rent for Ticket + Core asset is NOT yet measured: measure it on devnet in build step B4b before setting the ticket price. The chain-facts report quotes 0.00128 and 0.004 SOL for the same item.

LEGEND

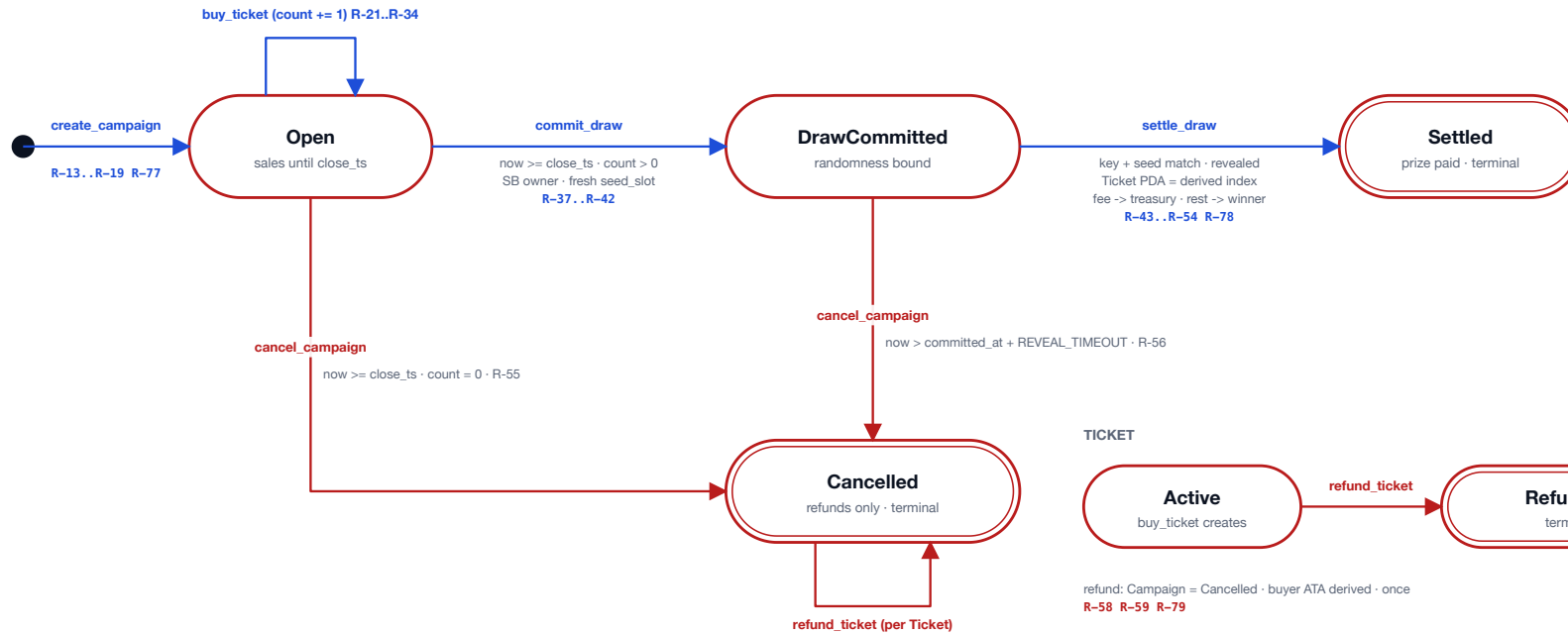
Actor Client (browser) Off-chain service Instruction Program-owned account External / third-party State machine Deferred (post-MVP) → call / CPI → optional / read / off-chain R-xx = requirement id

State machines

D3 — Campaign and Ticket state machines

Every arrow names its single owning instruction and guard. Settled and Cancelled are absorbing. Pause is a Config flag, not a state.

CAMPAIGN

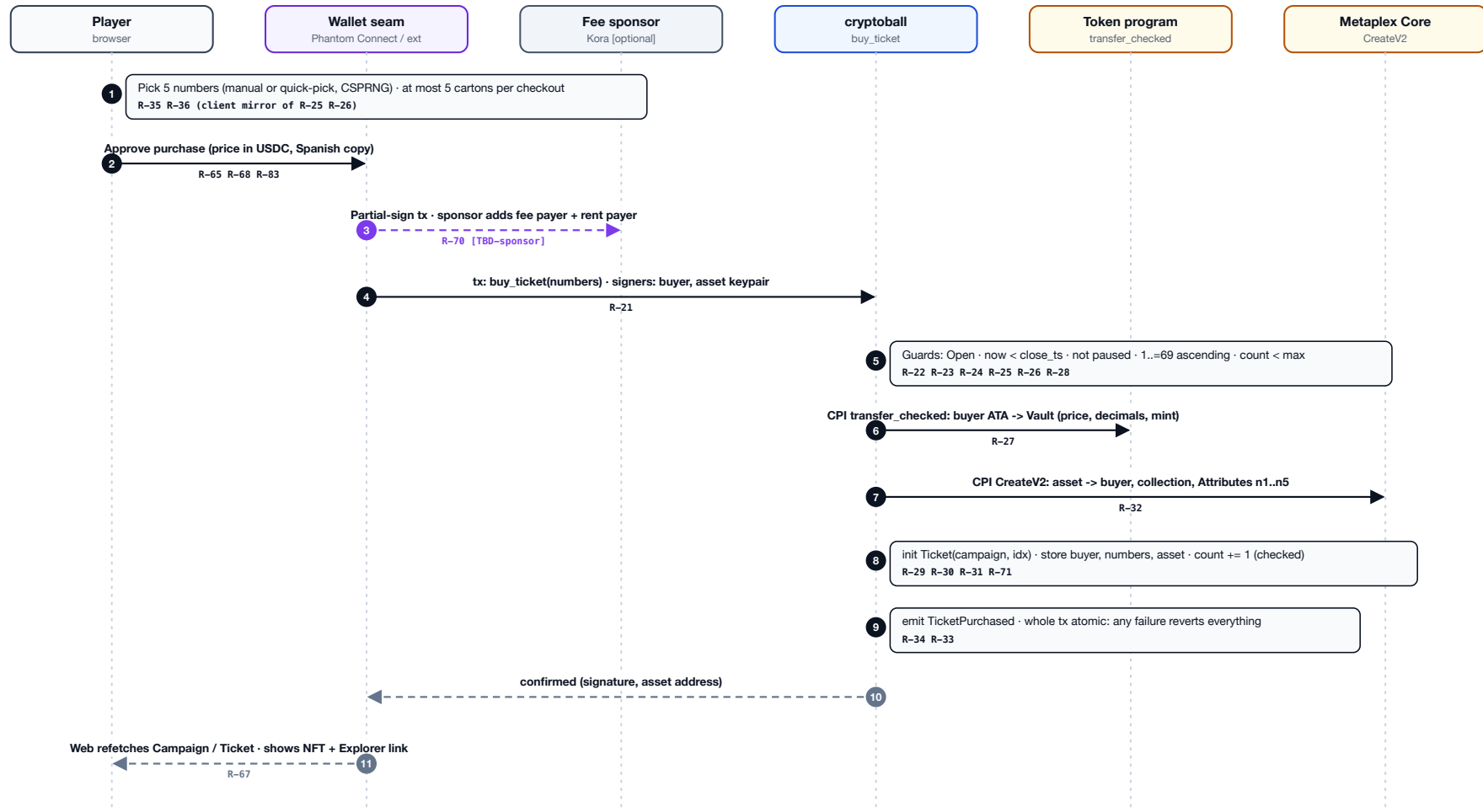


A Settled campaign's tickets stay Active: only the one winner is paid (default D2). Config.paused blocks buy_ticket only; commit, settle, cancel and refund always work, so funds are never trapped (R-24, R-87).

Sequence — buy_ticket

D4 — buy_ticket: Ticket absent → Active, Vault credit, asset mint

Owner of this state change: buy_ticket.

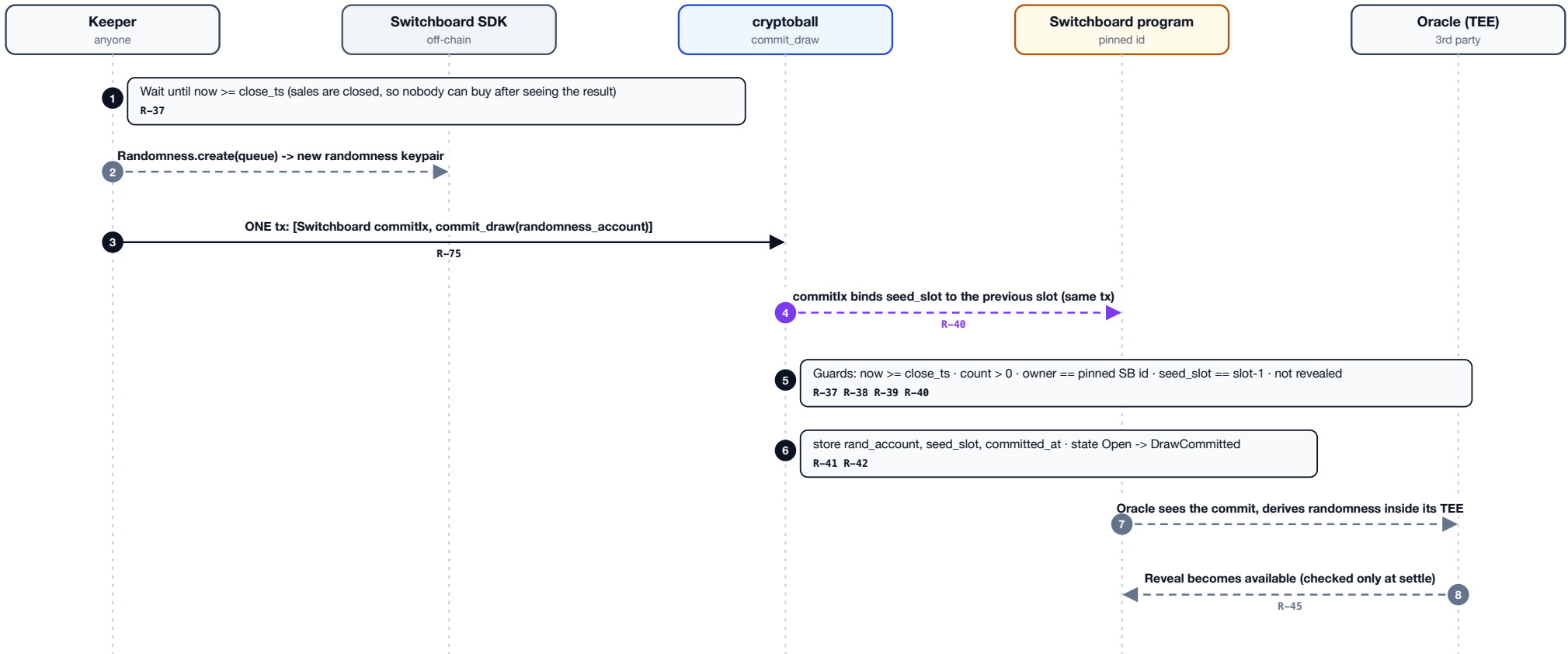


Failure paths (each has a negative test): closed, paused, after close, number out of range, unsorted/duplicate, sold out, insufficient USDC, wrong mint, wrong Core program id, forced CPI failure -> vault and count unchanged. No randomness is touched at purchase.

Sequence — commit_draw

D5 — commit_draw: Open → DrawCommitted

Owner: commit_draw. One commit per campaign; there is no re-commit instruction (no re-roll).

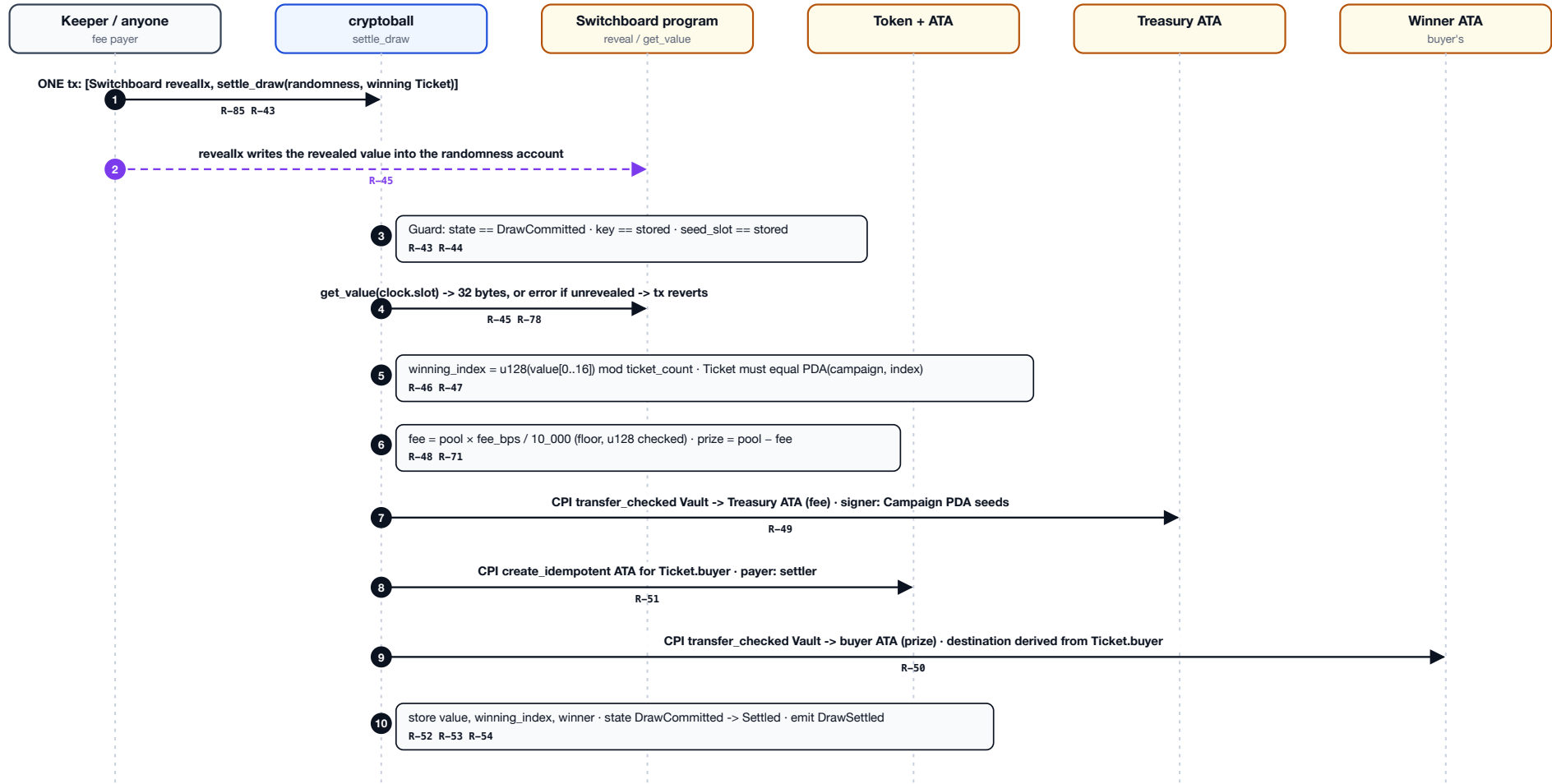


Residual risk (accepted for MVP, documented): the committer can see the revealed value off-chain and decline to settle. Settle is permissionless and `cancel_campaign` refunds everyone after `REVEAL_TIMEOUT`, so aborting costs a draw but gains nothing; there is no re-commit path. Switchboard guidance: take collateral at commit time (here: at purchase).

Sequence — settle_draw

D6 — settle_draw: DrawCommitted → Settled with automatic payout

Owner: settle_draw. Nobody claims; the winner signs nothing; the caller chooses no destination.



Modulo bias: reducing a uniform 128-bit value mod N ($N \leq 2^{32}$) biases by at most $N/2^{128}$ ($\sim 1e-29$): negligible, no rejection loop needed. The winner rule is the swappable unit [TBD-prize-tiers]: under default D2 exactly one ticket wins, so one tx pays everything. Exact-match tiers need a tally pass and per-winner payouts: a different instruction set.

Reference tables

State transition owners

State transition	Owning instruction
Config absent -> present	initialize
Config.fee/treasury/paused changes	update_config
Config.pending_admin set	nominate_admin
Config.admin rotated	accept_admin
Campaign absent -> Open	create_campaign
Ticket absent -> Active (+vault credit, +asset)	buy_ticket
Campaign Open -> DrawCommitted	commit_draw
Campaign DrawCommitted -> Settled (+fee, +prize)	settle_draw
Campaign Open DrawCommitted -> Cancelled	cancel_campaign
Ticket Active -> Refunded (+vault debit)	refund_ticket

Elements → requirements

ID	Element	Kind	Seeds / signer	Requirements
A-PLAYER	Player / buyer	actor	signs buy_ticket only; never signs payout	R-21, R-65
A-ADMIN	Operator / admin	actor	signs admin instructions	R-07, R-08, R-09, R-10, R-11, R-13
A-NOMINEE	Nominated admin	actor	signs accept_admin	R-12
A-UPGRADE	Upgrade authority	actor	signs initialize only	R-02
A-KEEPER	Keeper / anyone	actor	fee payer only; chooses no destination	R-37, R-43, R-55, R-56, R-58
A-TREASURY	Treasury (stakeholder)	actor	does not sign; address pinned in Config	R-03, R-49
A-ORACLE	Switchboard oracle (TEE)	actor	3rd party; writes randomness account	R-40, R-45
C-WEB	Web app (Next.js)	client		R-35, R-36, R-62, R-63, R-64, R-66, R-67, R-81, R-82, R-84
C-I18N	i18n: es-PE / es-419	client		R-68, R-69, R-83
C-WALLET	Wallet seam	client		R-65
C-KEEPER	Keeper script (tsx)	service		R-75, R-85, R-86
C-SPONSOR	Fee sponsor (Kora) [optional]	service		R-70
C-RPC	RPC + DAS read	service		R-61, R-62, R-63
C-META	NFT metadata (static)	service		R-32
DEF-ONRAMP	Fiat on-ramp (post-MVP)	deferred		deferred (no MVP requirement)
I-INIT	initialize	ix		R-01, R-02, R-03, R-04, R-05, R-06, R-76
I-UPDATE	update_config	ix		R-06, R-07, R-08, R-09, R-10, R-76
I-NOMINATE	nominate_admin	ix		R-11, R-76
I-ACCEPT	accept_admin	ix		R-12, R-76
I-CREATE	create_campaign	ix		R-13, R-14, R-15, R-16, R-17, R-18, R-19, R-77
I-BUY	buy_ticket	ix		R-21, R-22, R-23, R-24, R-25, R-26, R-27, R-28, R-29, R-30, R-31, R-32, R-33, R-34, R-70

ID	Element	Kind	Seeds / signer	Requirements
I-COMMIT	commit_draw	ix		R-37, R-38, R-39, R-40, R-41, R-42
I-SETTLE	settle_draw	ix		R-43, R-44, R-45, R-46, R-47, R-48, R-49, R-50, R-51, R-52, R-53, R-54, R-78
I-CANCEL	cancel_campaign	ix		R-55, R-56, R-60
I-REFUND	refund_ticket	ix		R-58, R-59, R-79, R-80
P-CONFIG	Config	pda	["config"]	R-01, R-03, R-04, R-05, R-61
P-CAMPAIGN	Campaign	pda	["campaign", id u64 LE]	R-19, R-20, R-41, R-52, R-61, R-77
P-TICKET	Ticket	pda	["ticket", campaign, index u32 LE]	R-29, R-30, R-59, R-61
P-VAULT	Vault (ATA of Campaign)	pda	ATA(campaign PDA, mint)	R-17, R-27, R-61
X-COLLECTION	Core collection (per campaign)	ext		R-18
X-ASSET	Core asset = ticket NFT	ext		R-32
X-RANDOM	Randomness account	ext		R-39, R-40, R-41, R-44, R-45, R-78
X-TREASURYATA	Treasury token account	ext		R-49
X-SB	Switchboard On-Demand	ext		R-73
X-CORE	Metaplex Core	ext		R-73
X-TOKEN	Token + ATA programs	ext		R-51, R-73
S-CAMPAIGN	Campaign state machine	state		R-42, R-53, R-57, R-87
S-GLOBAL	Cross-cutting rules (NFR)	state		R-71, R-72, R-73, R-74

Traceability

Requirement → elements, with negative test

Req	The protocol shall ...	Elements	Negative test
R-01	The protocol shall create one global Config account (seed "config") on initialize.	I-INIT , P-CONFIG	2nd initialize fails
R-02	The protocol shall reject initialize unless it is signed by the program upgrade authority.	A-UPGRADE , I-INIT	random signer initialize fails
R-03	The protocol shall store the treasury wallet address in Config at initialize.	A-TREASURY , I-INIT , P-CONFIG	state assertion
R-04	The protocol shall store the entry-token mint (USDC) in Config at initialize.	I-INIT , P-CONFIG	state assertion
R-05	The protocol shall store the fee in basis points in Config at initialize.	I-INIT , P-CONFIG	state assertion
R-06	The protocol shall reject any fee above MAX_FEE_BPS whenever the fee is set.	I-INIT , I-UPDATE	fee = MAX+1 fails (init and update)
R-07	The protocol shall allow the admin to update the fee.	A-ADMIN , I-UPDATE	non-admin update fails
R-08	The protocol shall allow the admin to update the treasury address.	A-ADMIN , I-UPDATE	non-admin update fails
R-09	The protocol shall allow the admin to pause ticket sales.	A-ADMIN , I-UPDATE	non-admin pause fails
R-10	The protocol shall allow the admin to resume ticket sales.	A-ADMIN , I-UPDATE	non-admin resume fails
R-11	The protocol shall allow the admin to nominate a new admin.	A-ADMIN , I-NOMINATE	non-admin nominate fails
R-12	The protocol shall require the nominee's signature to accept the admin role.	A-NOMINEE , I-ACCEPT	accept without nominee signature fails
R-13	The protocol shall allow the admin to create a campaign with an id, ticket price, close timestamp and max tickets.	A-ADMIN , I-CREATE	non-admin create fails
R-14	The protocol shall reject campaign creation when the close timestamp is not in the future.	I-CREATE	close_ts <= now fails
R-15	The protocol shall reject campaign creation when the ticket price is zero.	I-CREATE	price 0 fails
R-16	The protocol shall reject campaign creation when max tickets is zero or above MAX_TICKETS.	I-CREATE	max 0 and MAX+1 fail
R-17	The protocol shall create the campaign vault as the associated token account of the Campaign PDA for the configured mint.	I-CREATE , P-VAULT	vault authority/mint asserted
R-18	The protocol shall create one Metaplex Core collection per campaign via CPI.	I-CREATE , X-COLLECTION	wrong Core program id fails
R-19	The protocol shall initialize the campaign in state Open with ticket count zero.	I-CREATE , P-CAMPAIGN	state assertion
R-20	The protocol shall expose no instruction that changes a campaign's price, fee or close time after creation.	P-CAMPAIGN	grep: no handler mutates price/fee/close_ts
R-21	The protocol shall allow a buyer to purchase one ticket by supplying exactly five numbers.	A-PLAYER , I-BUY	happy path
R-22	The protocol shall reject a purchase when the campaign is not Open.	I-BUY	buy after DrawCommitted fails
R-23	The protocol shall reject a purchase at or after the close timestamp.	I-BUY	buy at close_ts fails
R-24	The protocol shall reject a purchase while sales are paused.	I-BUY	buy while paused fails

Req	The protocol shall ...	Elements	Negative test
R-25	The protocol shall reject any picked number outside 1..=69.	I-BUY	0 and 70 fail
R-26	The protocol shall reject picked numbers that are not strictly ascending.	I-BUY	[5,5,...] and [9,3,...] fail
R-27	The protocol shall transfer the ticket price in USDC from the buyer to the campaign vault using transfer_checked.	I-BUY , P-VAULT	insufficient USDC / wrong mint / wrong decimals fail
R-28	The protocol shall reject a purchase when the ticket count equals max tickets.	I-BUY	ticket max+1 fails
R-29	The protocol shall create a Ticket account derived from the campaign and ticket index.	I-BUY , P-TICKET	reused index fails
R-30	The protocol shall store buyer, numbers, index and Core asset address in the Ticket.	I-BUY , P-TICKET	state assertion
R-31	The protocol shall increment the campaign ticket count with checked arithmetic.	I-BUY	count near u32::MAX fails
R-32	The protocol shall mint one Core asset to the buyer with the numbers as an Attributes plugin.	C-META , I-BUY , X-ASSET	wrong Core id fails; attributes asserted
R-33	The protocol shall complete payment, Ticket creation and asset mint atomically in one transaction.	I-BUY	forced CPI failure -> vault+count unchanged
R-34	The protocol shall emit a TicketPurchased event.	I-BUY	parse event
R-35	The client shall generate quick-pick numbers from a cryptographically secure random source in ascending order.	C-WEB	UI test
R-36	The client shall allow at most five cartons per checkout.	C-WEB	UI test
R-37	The protocol shall allow anyone to commit the draw only at or after the close timestamp.	A-KEEPER , I-COMMIT	commit before close_ts fails
R-38	The protocol shall reject commit_draw when the campaign has zero tickets.	I-COMMIT	commit with 0 tickets fails
R-39	The protocol shall reject a randomness account not owned by the Switchboard On-Demand program id pinned for the cluster.	I-COMMIT , X-RANDOM	fake randomness account (right size) fails
R-40	The protocol shall reject a randomness account whose seed slot is not the previous slot or that is already revealed.	A-ORACLE , I-COMMIT , X-RANDOM	stale seed_slot / already revealed fails
R-41	The protocol shall store the randomness account address and seed slot in the Campaign.	I-COMMIT , P-CAMPAIGN , X-RANDOM	state assertion
R-42	The protocol shall move the Campaign from Open to DrawCommitted on commit_draw.	I-COMMIT , S-CAMPAIGN	commit twice fails
R-43	The protocol shall allow anyone to settle only when the Campaign is DrawCommitted.	A-KEEPER , I-SETTLE	settle before commit fails; settle twice fails
R-44	The protocol shall reject settlement when the randomness account or seed slot differs from the stored values.	I-SETTLE , X-RANDOM	swapped randomness account fails
R-45	The protocol shall obtain the 32-byte revealed value from the randomness account.	A-ORACLE , I-SETTLE , X-RANDOM	state assertion with known vector
R-46	The protocol shall derive the winning ticket index as the revealed value (u128) modulo the ticket count.	I-SETTLE	known value -> known index; distribution test
R-47	The protocol shall reject settlement unless the supplied Ticket account is the PDA for the derived index.	I-SETTLE	wrong Ticket account fails
R-48	The protocol shall compute the fee as pool times fee bps over 10,000, rounded down, with checked u128 math.	I-SETTLE	overflow price*count fails; rounding vector
R-49	The protocol shall transfer the fee from the vault to the treasury token account derived from Config.	A-TREASURY , I-SETTLE , X-TREASURYATA	wrong treasury token account fails
R-50	The protocol shall transfer the remaining pool from the vault to the winning buyer's token account.	I-SETTLE	destination != Ticket.buyer ATA fails

Req	The protocol shall ...	Elements	Negative test
R-51	The protocol shall create the winning buyer's token account if it does not exist, paid by the settler.	I-SETTLE , X-TOKEN	missing winner ATA is created, settler pays
R-52	The protocol shall store the revealed value, winning index and winner in the Campaign.	I-SETTLE , P-CAMPAIGN	state assertion
R-53	The protocol shall move the Campaign from DrawCommitted to Settled exactly once.	I-SETTLE , S-CAMPAIGN	second settle fails
R-54	The protocol shall emit a DrawSettled event.	I-SETTLE	parse event
R-55	The protocol shall allow anyone to cancel an Open campaign with zero tickets after its close timestamp.	A-KEEPER , I-CANCEL	cancel with tickets (not stuck) fails
R-56	The protocol shall allow anyone to cancel a DrawCommitted campaign after the reveal timeout has elapsed.	A-KEEPER , I-CANCEL	cancel before timeout fails
R-57	The protocol shall treat Settled and Cancelled as absorbing states.	S-CAMPAIGN	every ix on terminal state fails except refund
R-58	The protocol shall allow anyone to refund one Ticket's price to its stored buyer on a Cancelled campaign.	A-KEEPER , I-REFUND	refund on Settled/Open fails; wrong destination fails
R-59	The protocol shall mark a refunded Ticket Refunded.	I-REFUND , P-TICKET	state assertion
R-60	The protocol shall emit an event on campaign cancellation.	I-CANCEL	parse event
R-61	The protocol shall keep campaign, ticket, winner, randomness value and vault balance readable from public accounts without our API.	C-RPC , P-CONFIG , P-CAMPAIGN , P-TICKET , P-VAULT	decode accounts with IDL only
R-62	The client shall list campaigns with their prize pool.	C-WEB , C-RPC	UI test
R-63	The client shall let a user look up tickets by wallet address or purchase signature.	C-WEB , C-RPC	UI test
R-64	The client shall show the winning ticket and its numbers on the results page.	C-WEB	UI test
R-65	The client shall connect a wallet through embedded social login or a wallet-adaptor extension.	A-PLAYER , C-WALLET	UI test
R-66	The client shall show a devnet banner.	C-WEB	UI test
R-67	The client shall refetch on-chain state after every confirmed transaction.	C-WEB	UI test
R-68	The client shall present all user-facing text in Spanish (es-PE with es-419 fallback).	C-I18N	UI test
R-69	The client shall map every program error code to a Spanish message.	C-I18N	UI test
R-70	[TBD-sponsor] The client shall let a buyer without SOL purchase by routing fees and rent through a sponsor fee payer.	C-SPONSOR , I-BUY	[TBD-sponsor] buyer with 0 SOL purchases on devnet
R-71	The protocol shall use checked arithmetic on every money and counter path.	S-GLOBAL	boundary values fail, never wrap
R-72	The protocol shall return typed errors instead of panicking on user-controlled input.	S-GLOBAL	fuzzed inputs never abort
R-73	The protocol shall pin the Switchboard (per cluster), Core, token and associated-token program ids and never read them from instruction accounts.	X-SB , X-CORE , X-TOKEN , S-GLOBAL	attacker program id fails
R-74	The system shall operate on devnet only in the MVP.	S-GLOBAL	n/a (process)
R-75	The keeper script shall call commit_draw once a campaign has closed.	C-KEEPER	UI test
R-76	The protocol shall emit an event on every Config mutation.	I-INIT , I-UPDATE , I-NOMINATE , I-ACCEPT	parse event on each mutation

Req	The protocol shall ...	Elements	Negative test
R-77	The protocol shall snapshot the Config fee into the Campaign at creation.	I-CREATE , P-CAMPAIGN	later Config fee change leaves campaign unchanged
R-78	The protocol shall reject settlement when the randomness has not yet been revealed.	I-SETTLE , X-RANDOM	settle before reveal fails
R-79	The protocol shall reject a second refund of the same Ticket.	I-REFUND	second refund fails
R-80	The protocol shall emit an event on ticket refund.	I-REFUND	parse event
R-81	The client shall show a countdown to each campaign's close time.	C-WEB	UI test
R-82	The client shall block transactions when the wallet cluster is not devnet.	C-WEB	UI test
R-83	The client shall format money, numbers and dates through Intl for the active Spanish locale.	C-I18N	UI test
R-84	The client shall link the results page to the randomness account on the Explorer.	C-WEB	UI test
R-85	The keeper script shall call settle_draw once the randomness is revealed.	C-KEEPER	UI test
R-86	The keeper script shall call cancel_campaign when a campaign is cancellable.	C-KEEPER	UI test
R-87	The protocol shall allow commit_draw, settle_draw, cancel_campaign and refund_ticket while sales are paused.	S-CAMPAIGN	commit/settle/cancel/refund succeed while paused

Threat model

Attacker controls every account passed to an instruction until validated.

Threat / attacker goal	Mitigation in design	Requirements	Negative test	Source
Last-actor advantage: buy a ticket after seeing the result	Sales close at close_ts; commit_draw only after close (R-37); randomness unknown until committed	R-23 R-37	buy at close_ts fails; commit before close fails	FYEO randomness-ordering
Fake randomness account (right size, attacker-owned)	owner == pinned Switchboard id per cluster; key and seed_slot bound at commit	R-39 R-44 R-73	fake account fails commit; swapped account fails settle	FYEO-BANGER-03
Advisory randomness: result replaced by an admin/caller value	Winner is computed inside settle_draw from the account value; no instruction accepts a winner	R-46 R-47 R-52	wrong Ticket account fails; no admin_set_winner exists	FYEO-WAT-07
Stale / re-used / partially used randomness; dual program ids	seed_slot == slot-1 at commit; one commit per campaign; full 128 bits used; id is a per-cluster constant	R-40 R-41 R-46 R-73	stale seed fails; second commit fails	FYEO-WAT-10
Selective reveal: committer aborts when the result is unfavourable	No re-commit; settle permissionless; cancel + full refund after REVEAL_TIMEOUT. Residual: grief, no gain	R-56 R-58	cancel before timeout fails; refund once	Switchboard tutorial (collateral at commit)
Treasury redirection	Treasury token account derived from Config; never passed by caller	R-49	wrong treasury account fails	FYEO-BANGER-02
First-come initialize / re-init	Only the upgrade authority may initialize; init (not init_if_needed)	R-01 R-02	2nd and non-authority initialize fail	FYEO initialization class
Admin mutates a live campaign (price, fee, close)	No such instruction; fee snapshotted	R-19 R-20 R-77	later Config change leaves campaign unchanged	FYEO-WAT-05
Unbounded winner / ticket iteration (DoS)	Winner found by PDA derivation from index; no loops over tickets	R-47 R-72	settle cost independent of ticket count	FYEO dos-resource
NFT owner != buyer: payout confusion or theft by buying the NFT	Payout reads Ticket.buyer; NFT is a receipt (UI says so)	R-50	destination != Ticket.buyer ATA fails	own analysis
Admin key compromise / upgrade authority	Two-step rotation; recorded as devnet limitation; multisig/timelock is a mainnet gate	R-11 R-12 R-76	accept without nominee fails	PRINCIPLES.md Part B
Token account tricks (wrong mint, delegate, owner)	mint == Config.mint, delegate None, owners derived; transfer_checked	R-27 R-49 R-50	wrong mint / delegated account fail	FYEO account-validation
Arithmetic: fee rounding, overflow	u128 checked math, floor; fee <= MAX_FEE_BPS	R-06 R-48 R-71	overflow vectors fail	PRINCIPLES.md Part B
Sponsor relayer abuse (drain fee payer)	Kora policy allowlists only buy_ticket/commit/settle; rate limits [TBD-sponsor]	R-70	sponsor refuses other programs	Kora README (rate limiting, allowlists)

Reviewer checklist — verification

Check	Result	Evidence
Requirements written first, each numbered, each one action/state change/CPI	PASS with 13 declared exceptions	87 numbered. First run of <code>check.py</code> flagged 19 compound statements; 9 were split (R-19, 45, 59, 60, 62, 64, 66, 68, 75); the 13 still flagged are each one account write or one property (listed in requirements-v0.2.md). The flag is a heuristic (' and '), so this is a human judgement the captain should confirm.
Every requirement → a diagram element, every element → a requirement	PASS (machine-checked)	<code>src/check.py</code> : 87/87 requirements map to ≥1 element; 37/37 MVP elements map to ≥1 requirement; 1 deferred element exempt and labelled.
All actors enumerated with signer status explicit	PASS	7 actors (direct, admin, nominee, upgrade authority, anyone/keeper, treasury stakeholder, 3rd-party oracle); each has a <code>sign</code> field, checked by script. Sponsor fee payer is a service, shown with its co-sign arrow.

Check	Result	Evidence
Each state change names one owning instruction (one sentence)	PASS (machine-checked)	Owner table below: 10 transitions, 10 instructions, each instruction owns ≥ 1 ; D3 arrows carry the same names.
State ownership marked (PDA vs off-chain) and PDA seeds shown	PASS	D2 and D1 show seeds, owner program and mutators; off-chain state (Switchboard randomness, Core assets, metadata) marked third-party.
One reference table: accounts, PDAs, instructions, external programs $\rightarrow$ requirements	PASS	Reference table below.
Consistent arrow direction; distinct boundaries for validation/external; consistent labels	PASS (visual review)	Calls/CPI solid, optional/read dashed; external programs sit in an orange trust boundary; all labels generated from one model. Visual check performed on rendered PNGs of D1–D6.
Non-MVP requirements cut	PASS	Deferred list in the report; deferred element drawn dashed with no requirement; unused standards (cNFT, Token-2022 ext, Arcium) boxed 'Not used'.
Self-check done $\rightarrow$ AI red-team evaluated $\rightarrow$ override decisions logged	OPEN	Self-check done. Red-team and the captain's override log are the gate G2 step; findings from my own pre-review are in research-and-decisions.md §6.
One diagram per state change	PASS	buy (D4), commit (D5), settle (D6), cancel (D7), refund (D8), config/authority (D9); create_campaign is the entry arrow in D3 and is not cross-program beyond one Core CPI shown in D1/D2.

Open decisions and defaults

#	Decision	Default drawn	Alternatives / needs
1	Prize rule	(C) one drawn ticket wins; picked numbers recorded only	(A) exact-match jackpot · (B) match-count tiers (needs a tally pass and solvency design)
2	Pick format	5 numbers from 1–69, no bonus ball, ≤ 5 cartons per checkout	PRD says “5/6 cartons”; confirm 5 or 6
3	Entry asset	USDC (stable price)	SOL
4	Economics	price, fee bps, max tickets, close time, reveal timeout are admin parameters	captain supplies values; hard caps are constants
5	Fee sponsor	optional (Kora) so a buyer needs no SOL; devnet falls back to faucet SOL	[TBD-sponsor] spike first
6	Funding rails after devnet	none in MVP	Privy + Meld, exchange P2P, local partner; Peru availability unverified
7	Spanish scope	es-PE primary, es-419 fallback	Peru only vs Peru + Central America; native reviewer
8	Winner-rule / entry rail / wallet vendor	Phantom Connect behind a wallet seam	Privy

Gate G2.5 – devnet spike before the build: prove Switchboard create/commit/reveal, Core CreateV2 + Attributes under Anchor 0.32.1, measured transaction size and compute for buy_ticket and settle_draw, and Kora fee/rent sponsorship. Rent per ticket and all costs are unmeasured until then.

Regulation is treated as out of scope for the target market per the captain; recorded as an accepted risk. Mainnet is not part of this document.